Intrusion Detection on NSL-KDD via Cost-Sensitive Learning and Imbalance-Aware Training

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**Selected category: Category 3.*

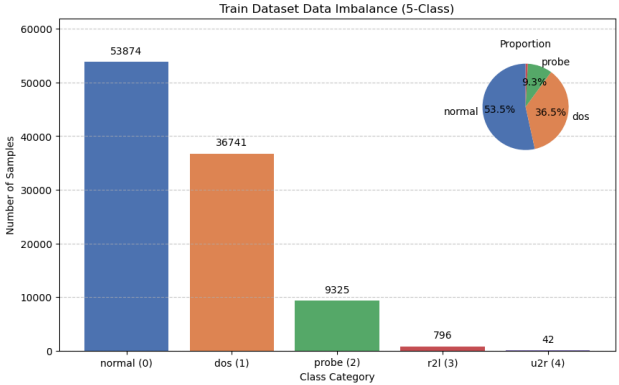
Background & Dataset Introduction

- What is an intrusion detection system (IDS)?
  - Monitors network traffic
  - Detects malicious activities
- NSL-KDD dataset
  - From KDD Cup 1999
  - Used for evaluating intrusion detection systems
- Attack categories
  - DoS: Exhaust system resources
  - Probe: Information gathering attacks
  - R2L: Remote access to local account
  - U2R: Privilege escalation attacks

Class Distribution

Data Characteristics

- 41 features:
  - 3 categorical
  - 38 numeric



Key Challenges

- Wide variation in scale and distribution
- Minority classes are extremely sparse

Problem & Motivation

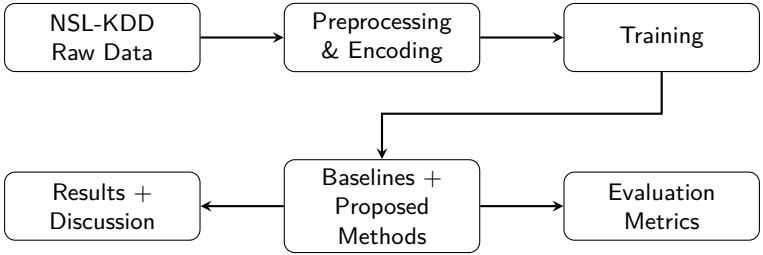
- IDS models must detect attacks under severe class imbalance.
- In NSL-KDD, rare classes (**R2L**, **U2R**) are low-frequency but high-impact.
- Accuracy-oriented training can hide critical minority-class failures.

Core Problem

How can we improve minority-attack detection reliability without introducing unacceptable false positives?

1. **RQ1:** How do common baselines behave on Macro-F1 and minority-class recall?
2. **RQ2:** Can imbalance-aware objectives improve minority detection with controlled false alarms?
3. **RQ3:** What errors remain, and what do they imply for IDS deployment?

Overall Experimental Pipeline



- Standard split: **KDDTrain+** for training/validation, **KDDTest+** for final evaluation.
- The pipeline is designed to support reproducibility, controlled comparison, and minority-class analysis.

Methods Compared in This Study

Baselines

- Logistic Regression
- Random Forest
- XGBoost
- MLP + standard Cross-Entropy

Proposed Imbalance-Aware Methods

- Logistic + weighted Cross-Entropy
- logistic + Focal Loss

Technical Focus

The central comparison is not merely between models, but between **standard objectives** and **objectives explicitly designed for imbalanced learning**.

Why Standard Objectives Are Insufficient

- Standard cross-entropy is dominated by the aggregate contribution of frequent classes.
- Under severe class imbalance, optimization can converge toward solutions that favor majority-class performance.
- Consequently, frequent and easy examples may overwhelm the learning signal from rare and difficult attacks.

Consequence

A model may appear effective under global metrics while still underperforming on **R2L** and **U2R**, which are the most operationally sensitive categories in this study.

Imbalance-Aware Training Objectives

Weighted Cross-Entropy

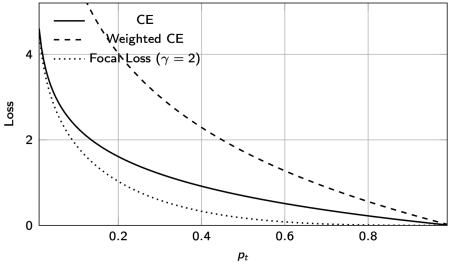
$$\mathcal{L}_{\text{WCE}} = -w_y \log(p_y), \qquad w_c \propto \frac{1}{N_c}$$

- Penalizes minority-class mistakes more heavily.
- Shifts learning focus by class frequency.

Focal Loss

$$\mathcal{L}_{\text{FL}} = \underbrace{\alpha_t}_{\text{class weight}} \cdot \underbrace{(1-p_t)^{\gamma}}_{\text{difficulty modulator}} \cdot \underbrace{(-\log p_t)}_{\text{CE}}$$

- Down-weights easy samples and emphasizes hard samples.
- Focuses on sample difficulty, not only rarity.

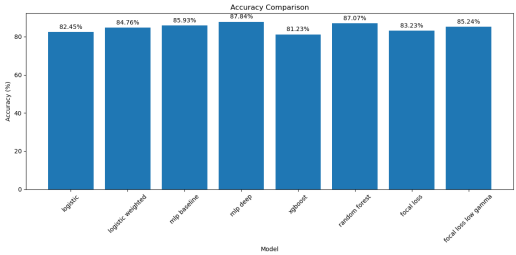


Focal example ($\gamma = 2$)

$p_t = 0.95 \Rightarrow (1-p_t)^2 = 0.0025$: easy sample receives little weight.
$p_t = 0.05 \Rightarrow (1-p_t)^2 = 0.9025$: hard sample receives strong weight.

Evaluation Design and Why Accuracy Is Not Enough

- Overall accuracy can stay high while R2L/U2R detection remains poor.
- Primary model-selection target: **Recall(R2L) + Recall(U2R)**.
- KDDTrain+ is split into train/validation sets (approximately 80/20).
- Early stopping rule: stop if $\text{Recall}_{\text{R2L}} + \text{Recall}_{\text{U2R}}$ does not improve for 5 rounds.
- The 5-round patience is empirical (practical heuristic, not theoretically unique).



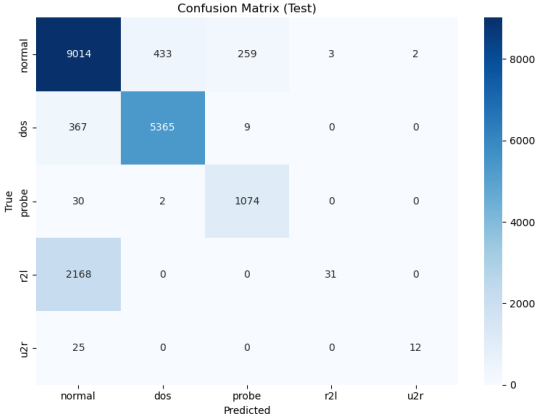
Accuracy remains high across all models, masking the failure to detect minority classes.

Alignment with project goal

Metrics and stopping criteria are intentionally chosen to improve minority-attack detection rather than optimize aggregate accuracy.

Baseline Logistic Regression: Multi-Class Confusion Matrix

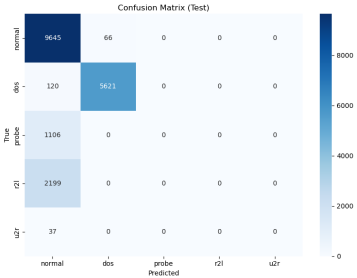
- Baseline logistic regression frequently maps minority attacks to **Normal**.
- This can preserve accuracy while creating high-risk missed detections.



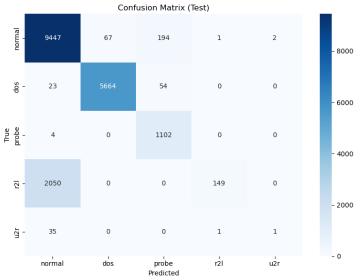
Other Baseline Models: XGBoost, Random Forest, and MLP

- XGBoost, Random Forest, and MLP improve majority-class classification compared with logistic regression.
- However, all still show substantial minority-class errors, especially attack-to-Normal confusions.

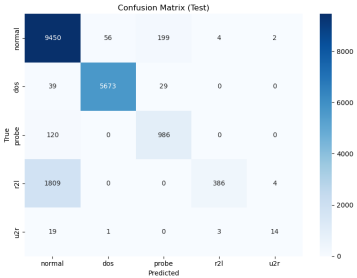
XGBoost



Random Forest



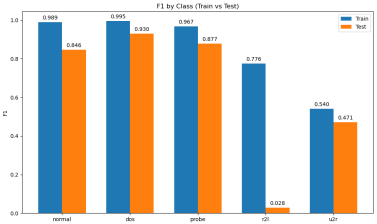
MLP



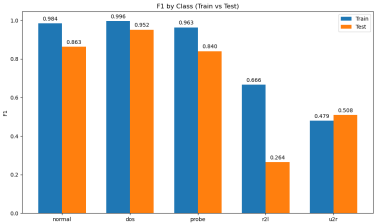
Multi-Class F1 Comparison: Logistic vs. Weighted vs. Focal

- Weighted loss and focal loss both improve minority-class F1 compared with baseline logistic regression.
- Gains differ by class, suggesting complementary behavior across optimization objectives.
- The next slide links these F1 gains to concrete confusion-matrix changes.

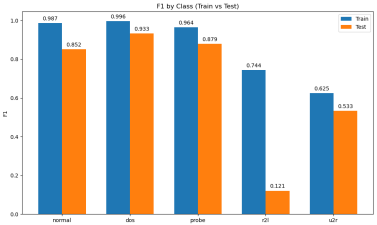
Baseline Logistic



Weighted Loss



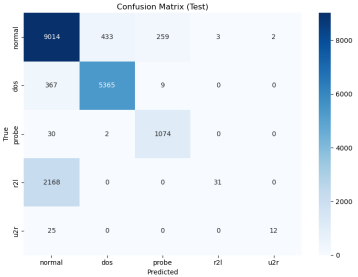
Focal Loss



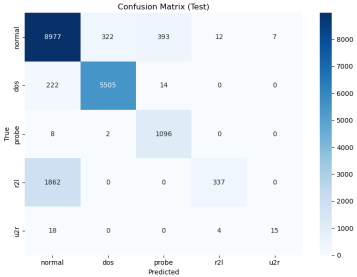
Confusion Matrix Comparison: Baseline vs. Weighted vs. Focal

- Loss redesign reduces **attack** → **Normal** errors across minority categories.
- Some false positives can increase, but missed dangerous attacks decrease.
- For IDS deployment, reducing false negatives on critical attacks is often the safer trade-off.

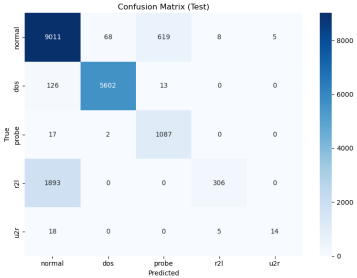
Baseline Logistic



Weighted Loss



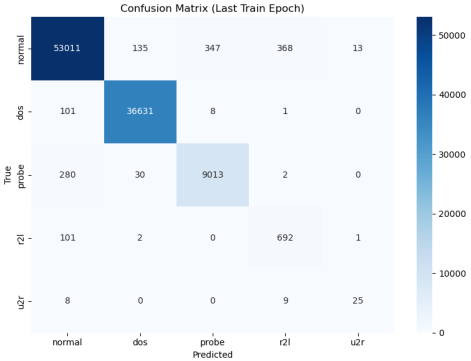
Focal Loss



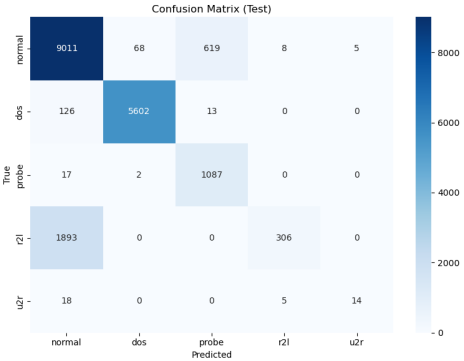
Why R2L Remains Difficult

- Even with improved losses, R2L remains the hardest class on the test set.
- Train performance is better than test performance, indicating limited generalization.
- Imbalance is only part of the issue; unseen R2L diversity also matters.

Train (Focal Loss)



Test (Focal Loss)



Test set contains significantly more R2L samples with unseen attack variants, suggesting the model has not learned the full R2L feature space.

Summary

- Standard models show a clear minority-class recall failure in imbalanced IDS classification.
- We redesign training objectives to target this weakness directly.
- Weighted loss and focal loss both improve minority-attack recognition.
- Improved models pay more attention to critical but underrepresented categories.
- In security practice, reducing missed attacks is often more important than minimizing false alarms.
- Our methods reduce attack-as-normal predictions, improving operational safety.
- R2L results indicate that loss redesign alone cannot fully solve generalization.
- More diverse R2L samples are likely required for stronger robustness.

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